NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 22/01/2026
Time: 3 Hrs | GTM – 22 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	A	D	D	D	B	C	A	C	B
11	12	13	14	15	16	17	18	19	20
B	B	C	D	B	B	B	A	D	C
21	22	23	24	25					
3	2	50	2	4					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	B	A	B	D	B	B	D	A	C
36	37	38	39	40	41	42	43	44	45
B	A	C	A	D	A	A	B	B	C
46	47	48	49	50					
22	4	75	47	4					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	A	C	C	A	D	C	B	D	B
61	62	63	64	65	66	67	68	69	70
C	B	C	D	D	B	B	C	A	A
71	72	73	74	75					
2187	3	16	490	182					

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HINTS
PHYSICS

1. Time dependent amplitude is a $\cos\left(\frac{\pi}{4}t\right)$

Intensity oscillates with angular frequency $2\left(\frac{\pi}{4}\right)=\frac{\pi}{2}$

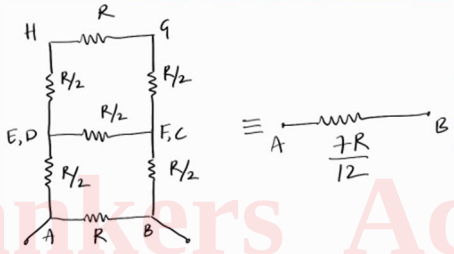
Time period $=\frac{2\pi}{\omega}=\frac{2\pi}{(\pi/2)}=4$

2. $A_{res}=\sqrt{A_1^2+A_2^2+2A_1A_2\cos(\Delta\theta)}$

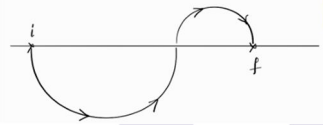
$=E_0\sqrt{1+4+2(1)(2)(1/2)}$

$=E_0\sqrt{3}$

3. If an external voltage is applied across the terminals A and B, terminals E and D will be equipotential similarly terminals F and C will be equipotential. The given network is equivalent to



4.



$S=2R_1+2R_2=2\left[\frac{mv}{qB_1}+\frac{mv}{qB_2}\right]$

$=\frac{2mv}{qB_1}\left[1+\frac{B_2}{B_1}\right]$

5. $\left[\frac{EJ^2}{M^5G^2}\right]=\frac{\left[ML^2T^{-2}\right]\left[ML^2T^{-1}\right]^2}{\left[M^5\right]\left[M^{-1}L^3T^{-2}\right]^2}=\left[M^0L^0T^0\right]$

$\vec{R}=\frac{m_1\vec{r}_1+m_2\vec{r}_2}{m_1+m_2}$

6. $=\frac{(3m)\left(\frac{3L}{2}\hat{i}\right)+(5m)\left(\frac{5L}{2}\hat{j}\right)}{3m+5m}$

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$$= \frac{L}{16}\left[9\hat{i} + 25\hat{j}\right]$$

7. CONCEPTUAL

$$\frac{1}{40} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{x} - \frac{1}{-x}\right)$$

$$\Rightarrow x = 40$$

8.

$$\frac{1}{f} = \left(\frac{3/2}{4/3} - 1\right)\left(\frac{1}{x}\right)$$

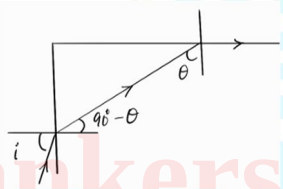
$$= \left(\frac{1}{8}\right)\left(\frac{1}{40}\right) \Rightarrow f = 320$$

9. $$V^2 = 16 + 49\cos^2\left(850\pi t + \frac{\pi}{8}\right) + 56\cos\left(850\pi t + \frac{\pi}{8}\right)$$

$$mean\left(v^2\right) = 16 + \frac{49}{2} + 0$$

$$\sqrt{mean\left(v^2\right)} = \frac{9}{\sqrt{2}}$$

10.



$$\mu \sin\theta = 1 \Rightarrow \theta = 60^0$$

$$\sin i = \mu \sin\left(90^0 - \theta\right)$$

$$= \frac{2}{\sqrt{3}} \times \frac{1}{2}$$

$$\Rightarrow i = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

11.

$$i_z = \frac{30-4}{500} - \frac{4}{200}$$

$$= 52mA - 20mA$$

$$= 32mA$$

12.

$$T_1 = \frac{2u\sin\theta}{g} \text{ and } T_2 = \frac{2u\cos\theta}{g}$$

$$T_1 T_2 = \frac{2u^2 \sin\left(2\theta\right)}{g^2} = \frac{2R}{g}$$

13. Conceptual

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14. $t_a = \sqrt{\dfrac{2\ell}{g(\sin\theta + \mu\cos\theta)}}$ and $t_d = \sqrt{\dfrac{2\ell}{g(\sin\theta - \mu\cos\theta)}}$

$t_d = \dfrac{140}{100} t_a$

$\Rightarrow \sin\theta + \mu\cos\theta = \dfrac{49}{25}(\sin\theta - \mu\cos\theta)$

$\Rightarrow \mu\cos\theta(25+49) = \sin\theta(49-25)$

$\Rightarrow \mu = \tan\theta\left[\dfrac{12}{37}\right]$

$\Rightarrow \mu = \dfrac{4\sqrt{3}}{37}$

15.

$\dfrac{1}{\lambda} = R\left(1 - \dfrac{1}{n^2}\right)$

$\Rightarrow \dfrac{1}{n^2} = 1 - \dfrac{1}{R\lambda}$

$\Rightarrow n = \sqrt{\dfrac{R\lambda}{R\lambda - 1}}$

16. $R \alpha n^2$ and $K.E \alpha \dfrac{1}{n^2}$

$\Rightarrow KR$ is independent of n

17. $f = \dfrac{qB}{2\pi m}$

f is independent of v

18.

$\vec{a} = \dfrac{\vec{F}}{m} = 8t^3\hat{i} - 6t^2\hat{j} - 2\hat{k}$

$\vec{v} = \int \vec{a}dt = 2t^4\hat{i} - 2t^3\hat{j} - 2t\hat{k}$

$P = \vec{F}.\vec{v} = \left(4t^3\hat{i} - 3t^2\hat{j} - \hat{k}\right)\left(2t^4\hat{i} - 2t^3\hat{j} - 2t\hat{k}\right)$

$= 2\left[4t^7 + 3t^5 + t\right]$

19.

$\sigma = y \in \Rightarrow \dfrac{F}{\dfrac{\pi d^2}{4}} = y\dfrac{e}{L}$

$\Rightarrow e = \dfrac{4FL}{\pi d^2 Y}$

$\Rightarrow \dfrac{e_1}{e_2} = \dfrac{L_1}{L_2}\left(\dfrac{d_2}{d_1}\right)^2 = \dfrac{4}{3}\left(\dfrac{5}{2}\right)^2 = \dfrac{25}{3}$

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20. $U_i = \left(\frac{-2kq^2}{2a}\right) + \left(\frac{-10kq^2}{3a}\right) + \left(\frac{5kq^2}{5a}\right) = \frac{-10kq^2}{3a}$

$U_f = \left(\frac{-2kq^2}{a}\right) + \left(\frac{-10kq^2}{a}\right) + \left(\frac{5kq^2}{a}\right) = \frac{-7kq^2}{a}$

$W = U_f - U_i = \frac{kq^2}{a}\left[\frac{10}{3} - 7\right] = \frac{-11kq^2}{3a}$

21. $I = \frac{ml^2}{12} + 2\left[\frac{ml^2}{12} + m\left(\frac{l}{2}\right)^2\right]$

$= \frac{ml^2}{4} + \frac{ml^2}{2} = \frac{3}{4}ml^2$

22.

$Z = \sqrt{R^2 + \left(X_L - X_c\right)^2} = 10\Omega$

$\Rightarrow i_{peak} = \frac{100\sqrt{2}}{10} = 10\sqrt{2}$

$\Rightarrow i_{rms} = 10$

Voltmeter reading is $i_{rms} X_c = 200$

23. Use apparent depth $= actual\ depth\left(\mu_2 / \mu_1\right)$

24.

$\frac{\sigma\left(4\pi R^2\right)T^4}{4\pi r^2} = S$

$\Rightarrow T = \left[\frac{S}{\sigma}\left(\frac{\gamma}{R}\right)^2\right]^{1/4}$

$\Rightarrow a = -2\ and\ b = \frac{1}{4}$

25.

$U_c = U_b \left[\because isothermal\right]$

$for\ bc,\ W + \Delta U = 0 \left[\because adiabatic\right]$

$\Rightarrow 4 + U_a - U_b = 0$

$\Rightarrow U_b - U_a = 4$

$\Rightarrow U_c - U_a = 4$

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CHEMISTRY

26. Secondary amines and Phenols give libermann nitroso test
27. Conceptual
28. Sucrose is an invert sugar
29.

Radiation Type	Gamma Ray	X-ray	Ultraviolet	Visible	Infrared	Microwave	Radio
Wavelength (m)	10^{-12}	10^{-10}	10^{-8}	5×10^{-6}	10^{-5}	10^{-1}	10^{3}

30. Conductivity varies with the nature of electrolyte as well as concentration
31. Conceptual
32. Vanadium has the highest enthalpy of atomization in 3d series. Its maximum oxidation number is +5
33. In first reaction base is sterically hindered thus alkene is produced via elimination
In second reaction alkene is produced due to iodination-deiodination
In 3rd and 4th cases dehydrobromination and debromination takes place respectively
34. $[H^+]_{net} = N_1V_1+N_2V_2+N_3V_3/(V_1+V_2+V_3$
35. $2.18\times10^6\times\frac{z}{n}=1.635\times10^6$

$2.18\times10^6\times\frac{z}{4}=1.635\times10^6$

$z=\frac{4\times1.635}{2.18}=3$

36. $HA\rightarrow H^+ + A^-;\ \Delta_r H = 1.4\,kJ/mol$

$\Delta H_{\text{neutralization}} = \Delta H_{\text{ionization}} + \Delta_r H$

$\left(H^+ + OH^- \rightarrow H_2O\right)$

$-55.95 = \Delta H_{\text{ionization}} - 57.3$

$\Delta H_{\text{ionization}}$ for $1MHA = 1.35\,kJ/mol$

% heat utilized by $1M$ acid for ionization

$=\frac{1.35}{1.4}\times100=96.43\%$

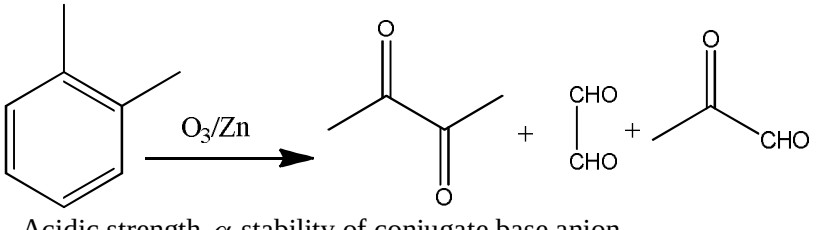
So, acid is $100-96.43=3.57\%$ ionized

$\left[H^+\right]=C\alpha=\left(1\times0.0357\right)M$

37. $SO_3^{2-}+H_2SO_4\rightarrow SO_2$

$SO_2+K_2Cr_2O_7\rightarrow Cr\left(SO_4\right)_{3aq}$

38.

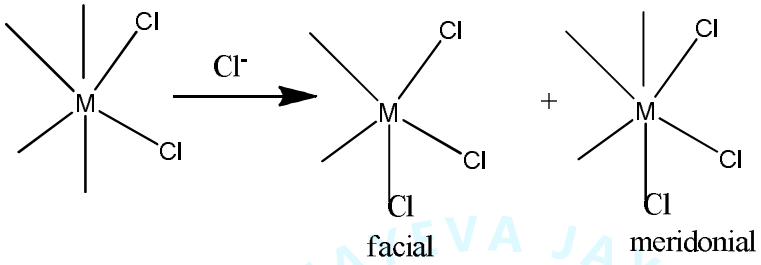


39. Acidic strength α stability of conjugate base anion

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40. Conc. of C increases gradually and decrease in conc. of B is twice that of A

42.



facial meridonial

43. Conceptual

44. Phenolic group cannot give $NaHCO_3$ test

45.

	A_4	$\longrightarrow$	$4A$
$t = 0$	aM		0
$t = 30$ min	$(a - x)$ M		$4x$M

As $a - x = 4x \Rightarrow x = \frac{a}{5}$

$\therefore$ Percentage reaction at $t = 30$ min

$$= \frac{x}{a} \times 100 = 20\%$$

Now, $30 = \frac{t_{1/2}}{\log 2} \cdot \log \frac{a}{a-x} \Rightarrow t_{1/2} = 90$ min

46. $\%carbon = \frac{12}{44} \times \frac{WCO_2}{W_{compound}} \times 100$

$= \frac{12}{44} \times \frac{0.198}{0.246} \times 100 = 22$

47. Conceptual

48. $i \times C = i \times C$

$i \times 0.004 = 1 \times 0.01$

$i = 2.5$

$1 + 2\alpha = 2.5$

$2\alpha = 1.5$

$\alpha = 0.75$

$\%\alpha = 75$

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50.

$$WO_2 = \frac{33.68}{100} \times 190$$

$$WO_2 = 64$$

$$n_{atoms} = \frac{64}{16} = 4$$

MATHEMATICS

51. Let E_1 be the event that one of first 'n' urns is chosen,

E_2 be the event that $(n+1)^{th}$ urn is chosen.

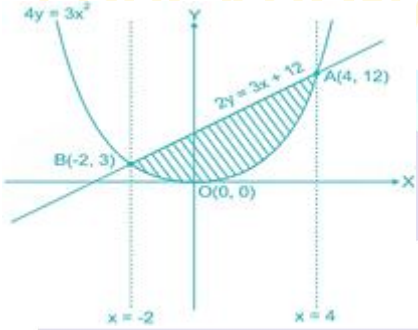
A be the event that two balls drawn are black.

$$P(E_1) = \frac{n}{n+1};\ P(E_2) = \frac{1}{n+1}$$

$$P\left(\frac{A}{E_1}\right) = \frac{{}^6C_2}{{}^{10}C_2} = \frac{1}{3};\ P\left(\frac{A}{E_2}\right) = \frac{{}^5C_2}{{}^{10}C_2} = \frac{2}{9}$$

$$P\left(\frac{E_2}{A}\right) = \frac{P(E_2).P\left(\frac{A}{E_2}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2).P\left(\frac{A}{E_2}\right)} \Rightarrow \frac{1}{16} = \frac{\frac{1}{n+1}.\frac{2}{9}}{\frac{n}{n+1}.\frac{1}{3} + \frac{1}{n+1}.\frac{2}{9}} \Rightarrow n = 10$$

52.



$$3x^2 = 2(3x+12)$$

$$3x^2 - 6x - 24 = 0$$

$$x^2 - 4x + 2x - 8 = 0$$

$$x = 4,\ x = -2$$

$$\int_{-2}^{4} \frac{3x+12}{2} - \frac{3x^2}{4}\,dx = A$$

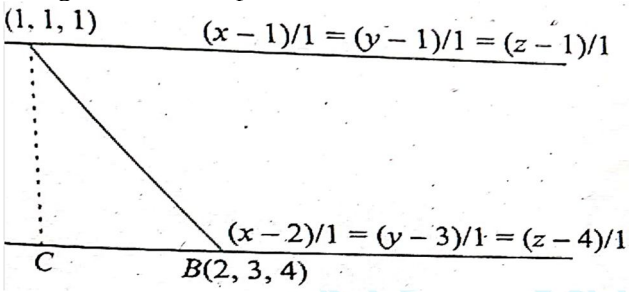
$$\left[\frac{3x^2}{4} + 6x - \frac{x^3}{4}\right]_{-2}^{4} = A$$

$$3\times 4 + 6\times 4 - \frac{4^3}{4} - \left[\frac{3}{4}.4 - 12 + \frac{8}{4}\right] = A$$

$$36 - 16 - [-9+2] = A \Rightarrow A = 27$$

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53. The given lines are parallel



From the figure, we get $BC+\frac{(2-1)1}{\sqrt{3}}+\frac{(3-1)1}{\sqrt{3}}+\frac{(4-1)1}{\sqrt{3}}=\frac{1+2+3}{\sqrt{3}}=2\sqrt{3}$

$AB=\sqrt{1+4+9}=\sqrt{14}$

Shortest distance $=AC=\sqrt{14-12}=\sqrt{2}$

54.

$f'(x)=\frac{a}{x}+2bx+1$

$f'(x)=0@x=1and\ x=3$

$\Rightarrow a+2b+1=0\Rightarrow(1)$

$\Rightarrow \frac{a}{3}+6b+1=0\Rightarrow(2)$

$\Rightarrow a+18b+3=0\rightarrow(3)$

$\Rightarrow -16b-2=0$

$b=\frac{-1}{8},a=\frac{-3}{4}$

55. $\vec{b}=2\vec{c}+\lambda\vec{a},\ \left|\vec{b}\right|^2=\left|2\vec{c}+\lambda\vec{a}\right|^2$

56. $a=5,ar^2=a+3d,ar^4=a+15d$

$5r^2=5+3d,5r^4=5+15d$

$r^4=1+3d$

$25r^4=25+75d$

$(5+3d)^2=25+75d$

$25+30d+9d^2=25+75d$

$9d^2-45d=0$

$d=5,0$

57. The given equation is

$x^2-2mx-1=0$

$\Rightarrow (x-m)^2-1=0$

$\Rightarrow (x-m+1)(x-m-1)=0$

$\Rightarrow x=m-1,m+1$

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From given condition,

$m-1>-2 and \;\; m+1<4$

$\Rightarrow m>-1 \; and \; m<3$

$-1<m<3$

58. We have

$32^{32}-2^{5\times 32}=2^{160}$

$=(3-1)^{160}=(3-1)^{160}$

$a\dfrac{32^{32}}{3}=\dfrac{1}{3}+\left[-C_1+C_2.3+C_3.3^2+.....+C_{160}.3^{159}\right]$

$=\dfrac{1}{3}+\text{int}\, eger$

Whose fractional part is 1/3

59. $e^{\lim\limits_{x\to 0}\frac{(\cos x-1)}{\sin^2 x}}+\lim\limits_{x\to 0}\dfrac{\dfrac{\sin 2x}{2x}.2x+6x.\dfrac{\tan^{-1}3x}{3x}+3x^2}{\dfrac{\ln\left(1+3x+\sin^2 x\right)}{3x+\sin^2 x}.\left(3x+\sin^2 x\right)+xe^x}=\dfrac{1}{\sqrt{e}}+2$

60.

$Let \;\; f(x)=y$

$\dfrac{dy}{dx}-2xy=\dfrac{e^{x^2}}{(x+1)^2}$

61. Case 1: Mr. 'B' is present 'A' is excluded &'C' included number of such committees

$={}^4C_2.{}^4C_1=24$

Case 2: Mr. 'B' is absent number of such committees = $={}^5C_3.{}^5C_2=100$

Total= 124

62. For three planes to intersect along line

$\begin{vmatrix} K & 4 & 1 \\ 4 & K & 2 \\ 2 & 2 & 1 \end{vmatrix}=0\Rightarrow K(K-4)+8-2K=0$

$\Rightarrow K^2-6x+8=0\Rightarrow K=24$

Determinant

$=\begin{vmatrix} 4 & 2 \\ 2 & 4 \end{vmatrix}=12$

$P^{-1}=\dfrac{1}{12}\begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$

63. A square matrix

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$n=3,|A|=2$

$|A^{-1}|=\frac{1}{|A|}$

$|adjA|=|A|^{n-1}$

Let $adjA^{-1}=B$

$\Rightarrow\left|\left(adjA^{-1}\right)^{-1}\right|=|B^{-1}|=\frac{1}{|B|}=\frac{1}{|adjA^{-1}|}$

$=\frac{1}{\left(|A^{-1}|\right)^{n-1}}=\frac{1}{\left(\frac{1}{|A|}\right)^{n-1}}=\frac{1}{\left(\frac{1}{2}\right)^{2}}=4$

64.

$I=\int_0^{\pi/2}\frac{x\sin x\cos x}{\cos^4 x+\sin^4 x}$

$I=\int_0^{\pi/2}\frac{(\pi/2-x)\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

Finding (1) and (2) we get

$I=\frac{\pi}{4}\int_0^{\pi/2}\frac{\sin x\cos x}{\sin^4 x+\cos^4}dx$

$=\frac{\pi}{4}\int_0^{\pi/2}\frac{\sec^2 x\tan x}{\tan^4 x+1}dx\left(Dividing\ \ Nr.\ and\ \ Dr.\ by\ \cos^4 x\right)$

$=\frac{\pi}{2\times 4}\int_0^{\pi/2}\frac{\left(\tan^2 x\right)dx}{1+\left(\tan^2 x\right)^2}$

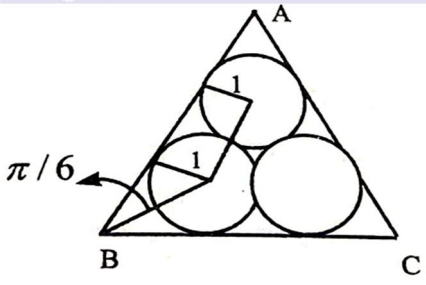
65. Equation of directrix is $x+y=0$

Hence equation of the parabola is $\frac{x+y}{\sqrt{2}}=\sqrt{(x-2)^2+(y-2)^2}$

Hence equation of parabola is $(x-y)^2=8(x+y-2)$

66. Length of a side $=2+2\cot\frac{\pi}{6}=2\left(1+\sqrt{3}\right)$

Area $=\frac{\sqrt{3}}{4}\left(2\left(1+\sqrt{3}\right)\right)^2=6+4\sqrt{3}$



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67. Orthocentre of the triangle ABC is (1,2)

Let coordinates of A are (h,k)

$$AD = \frac{k-2}{n-1}, slope\ BE = \frac{-5}{3}$$

$$AD \perp BC \Rightarrow \frac{k-2}{h-1} \times \left(\frac{8}{-6}\right) = -1$$

Slope $\Rightarrow 3h - 4k + 5 = 0$

$$Also\ BE \perp AC \Rightarrow \left(\frac{k-5}{h+2}\right)\left(\frac{-5}{3}\right) = -1$$

$$\Rightarrow 3h - 5k + 31 = 0 \Rightarrow A = (33, 26)$$

68. $\frac{\sum x_i}{20} = 10; \sum x_i = 10 \times 20 = 200$

If 8 is replaced by 12, then

$\sum x_i = 200 - 8 + 12 = 204$

Now, corrected mean $\left(\bar{x}\right) = \frac{\sum x_i}{20} = \frac{204}{20} = 10.2$

Standard deviation =2

So, Variance $= \left(S.D\right)^2 = 2^2 = 4$

By definition $\Rightarrow \frac{\sum x_i^2}{20} - \left(\frac{\sum x_i}{20}\right)^2 = 4 \Rightarrow \frac{\sum x_i^2}{20} - (10)^2 = 4$

$\Rightarrow \sum x_i^2 = 2080$

Now, replace observation '8' by '12'

Then $\sum x_1^2 = 2080 - 8^2 + 12^2 = 2160$

Variance of replaced observation Ps

$$\Rightarrow Var = \frac{\sum x_i^2}{20} - \left(\frac{\sum x_i}{20}\right)^{20} = \frac{2160}{20} - (10.2)^2$$

$= 108 - 104.04 = 3.96$

Corrected standard deviation $= \sqrt{Var} = \sqrt{3.96}$

69. $bx + ay - ab = 0$

$ax + by - 1 = 0$

$$x = \frac{a\left(b^2-1\right)}{b^2-a^2}, y = \frac{b\left(a^2-1\right)}{b^2-a^2}$$

$$x^2 + y^2 = \frac{a^2\left(b^2-1\right)^2 + b^2\left(a^2-1\right)^2}{\left(b^2-a^2\right)^2}$$

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$$=\frac{a^2b^2\left(a^2+b^2\right)-4a^2b^2+a^2+b^2}{\left(b^2-a^2\right)^2}$$

$$=\frac{a^3b^3-4a^2b^2+ab}{\left(b^2-a^2\right)^2}$$

$$=\frac{ab\left(a^2b^2-ab+1-3ab\right)}{\left(b^2-a^2\right)^2}$$

$$=\frac{ab\left(a^2b^2-ab+1\right)}{\left(b^2-a^2\right)^2}-\frac{3a^2b^2}{\left(b^2-a^2\right)^2}$$

$$-xy-\frac{3a^2b^2}{\left(b^2+a^2\right)^2-4a^2b^2}$$

$$-xy-\left(\frac{3a^2b^2}{-3a^2b^2}\right)$$

$$-xy+1=x^2+y^2$$

$$x^2+y^2+xy-1=0$$

70.

$$\Rightarrow\left|\frac{1+ia}{1-ia}\right|=|z|$$

$$\Rightarrow\left|\frac{a-i}{a+i}\right|^4=1$$

$$\Rightarrow|a-i|=|a\div i|$$

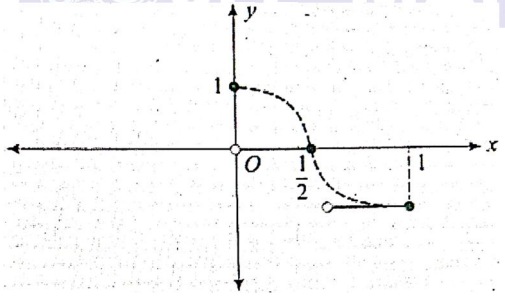
Therefore a lies on the perpendicular bisector of i and –i , which is real axis. Hence all the roots are real.

71. Trace $(A)=2\Rightarrow$ sum of diagonal elements =2, the possible are $(1,1,0),(0,1,1,),(1,0,1)$ and each of remaining places can be filled in 3 ways

Thus required number of ways $=3\times3^6=3^7=2187$

72. Consider $x\in[0,1]$

From the graph given, it is clear that $[\cos\pi x]$ is discontinuous at $x=0,1/2$



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Now consider $x \in (1,2]$

$$f(x) = [x-2][2x-3]$$

For $x \in (1,2), [x-2] = -1$ and for $x = 2, [x-2] = 0$

Also $|2x-3| = 0$ *or* $x = 3/2$

Therefore, $x = 3/2$ and 2 may be the points at which $f(x)$ is discontinuous

$$f(x) = \begin{cases} 1, & x = 0 \\ 0, & 0 < x \le \frac{1}{2} \\ -1, & \frac{1}{2} < x \le 1 \\ -(3-2x), & 1 < x \le 3/2 \\ -(2x-3), & 3/2 < x \le 2 \\ 0, & x = 2 \end{cases}$$

Thus $f(x)$ is continuous when $x \in [0,2] - \{0,12,2\}$

73.

$A = \{1,2,3,4\}$

$R = \{(1,2),(2,3),(1,4)\}$

S is equivalence for $R < S$ and reflexive

$\{(1,1),(2,2),(3,3),(4,4)\}$

For symmetric

$\{(2,1),(4,1),(3,2)\}$

For transitive

$\{(1,3,),(3,1),(4,2),(2,4)\}$

Now $setS = \{(1,1),(2,2),(3,3),(4,4),(1,2),(2,3),(1,4),(4,3),(3,4),(2,1),(4,1),(3,2),(1,3),(3,1),(4,2),(2,4)\}$

$n(S) = 16$

74.

$F(mn) = f(m).f(n)$

$put\ m = 1\ f(n) = f(1).f(n) \Rightarrow f(1) = 1$

$Put\ m = n = 2$

$$f(4) = f(2).f(2) \begin{cases} f(2) = 1 \Rightarrow f(4) = 1 \\ or \\ f(2) = 2 \Rightarrow f(4) = 4 \end{cases}$$

$put\ \ m = 2,\ n = 3$

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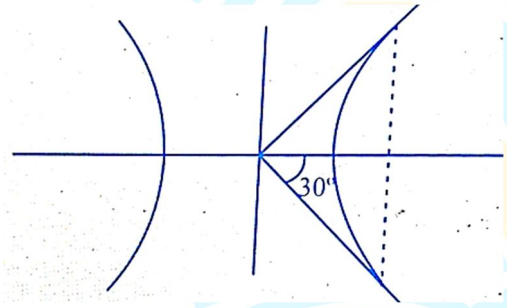
$$f(6)=f(2).f(3)\begin{cases}when\ f(2)=1\\ f(3)=1to7\\ f(2)=2\\ f(3)=1or2or3\end{cases}$$

f(5), f(7) can take any value

Total = $(1\times1\times7\times1\times7\times1\times7)+(1\times1\times3\times1\times7\times1\times7)$

= 490

75. LR subtends 60^0 at centre



$$\Rightarrow \tan 30^0=\frac{b^{2/a}}{ae}=\frac{b^2}{a^2e}=\frac{1}{\sqrt{3}}$$

$$\Rightarrow e=\frac{\sqrt{3}b^2}{9}$$

$$also\ e^2=1+\frac{b^2}{9}\Rightarrow 1+\frac{b^2}{9}=\frac{3b^4}{81}$$

Also

$$e^2=1+\frac{b^2}{9}\Rightarrow 1+\frac{b^2}{9}=\frac{3b^4}{81}$$

$$\Rightarrow b^4=3b^2+27$$

$$\Rightarrow b^4-3b^2-27=0$$

$$\Rightarrow b^2=\frac{3}{2}\left(1+\sqrt{13}\right)$$

$$\Rightarrow \ell=3, m=2, n=13$$

$$\Rightarrow \ell^2+m^2+n^2=182$$

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